Python eBPF: BCC C-String Runtime vs Python-BPF AST AOT

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About Me

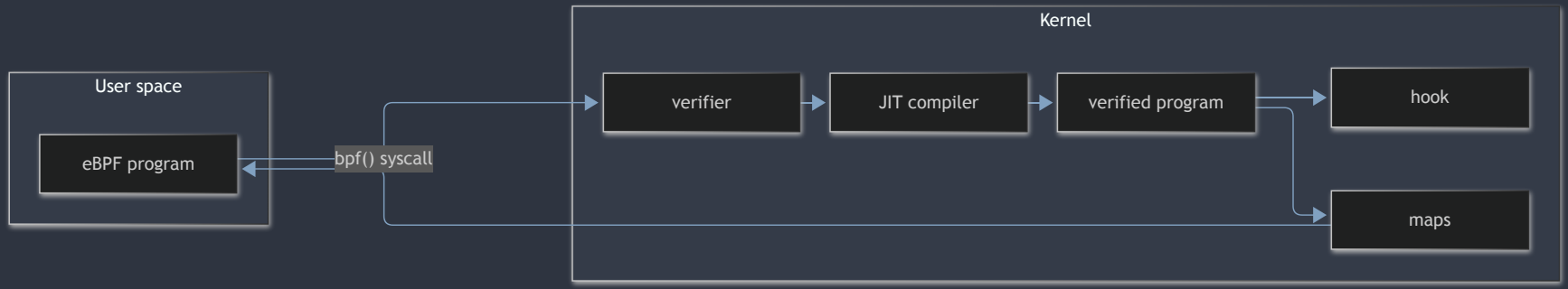
Systems engineer at DomaiNesia · Yogyakarta, Indonesia

- Focus: eBPF from Python — kernel observability without writing C
- Recently: Python wrappers for traffic filtering — utilizing bpfilter
- Today: PyCon ID 2026 · previously PyCon APAC 2024 · PyCon HK 2025 · PyCon ID 2025

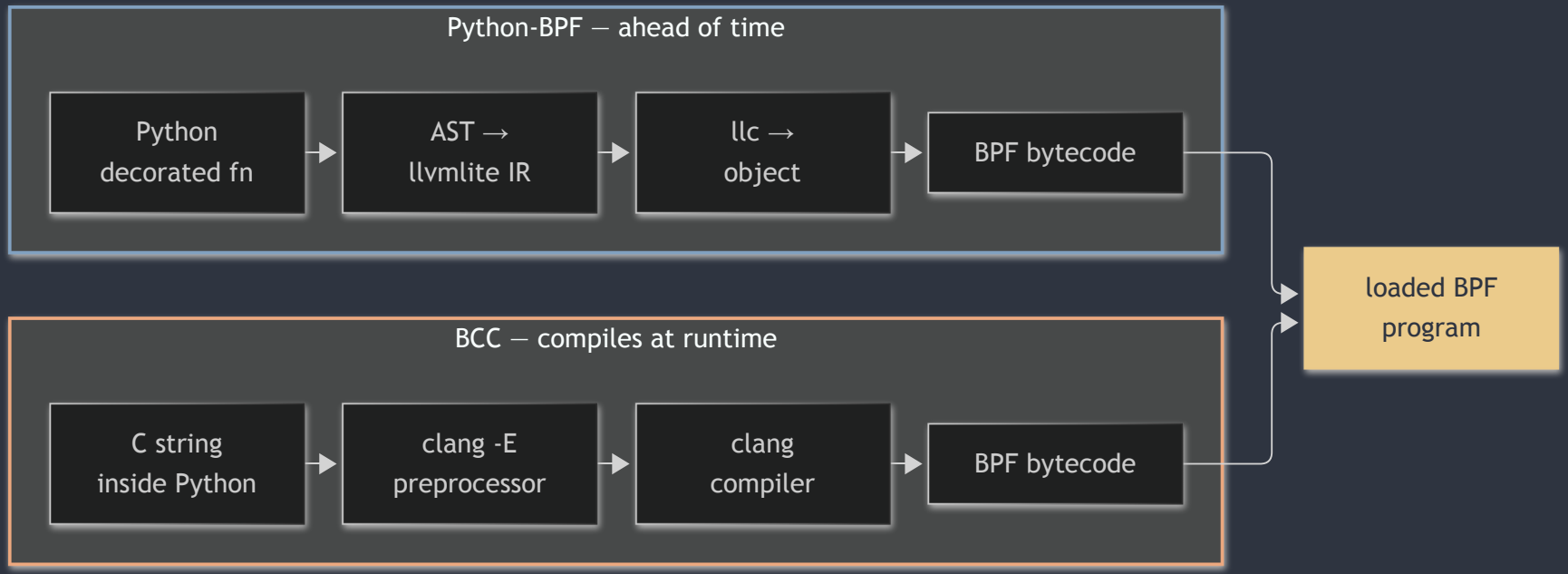
github.com/nizarakbar

talk repo

eBPF architecture overview



Architecture: How a Program Becomes Bytecode



Consequence: BCC needs `clang` + kernel headers on the host at runtime. Python-BPF needs neither once built.

Two frameworks, three questions

Where does state live? → maps

How does data reach userspace? → perf event array

How do you emit an event? → submit

State: the HashMap

BCC — C macro, compiled at runtime

```c
BPF_HASH(start, u32);   // key=u32, leaf=u64, size=10240

int probe(struct pt_regs *ctx) {
    u32 key = bpf_get_current_pid_tgid();
    u64 *val = start.lookup(&key);   // ptr to leaf
    if (val) start.update(&key, val);
    return 0;
}
```

Python-BPF — Python class, AOT

```python
@bpf
@map
def start() -> HashMap:
    return HashMap(key=c_uint32, value=c_uint64,
                   max_entries=10240)

@bpf
@section("tracepoint/...")
def probe(ctx):
    key = pid()
    val = start.lookup(key)          # value, not ptr
    if val: start.update(key, val)
    return 0
```

Userspace channel: the perf event array

BCC — `BPF_PERF_OUTPUT` + `perf_submit`

```
BPF_PERF_OUTPUT(events);

struct data_t { u32 pid; u64 ts; char comm[16]; };

int hello(struct pt_regs *ctx) {
    struct data_t d = {};
    d.pid = bpf_get_current_pid_tgid();
    d.ts  = bpf_ktime_get_ns();
    events.perf_submit(ctx, &d, sizeof(d));
    return 0;
}
```

Python-BPF — `PerfEventArray.output`

```
@bpf
@map
def events() -> PerfEventArray:
    return PerfEventArray(key_size=c_uint32,
                          value_size=c_uint32)

@bpf
@section("tracepoint/...")
def hello(ctx):
    d = ProcessEvent()
    d.pid = pid()
    d.ts  = ktime()
    events.output(d)     # sizeof() inferred
    return 0
```

Same job, different spelling

step	BCC	Python-BPF
declare map	`BPF_HASH(start, u32)`	`HashMap(key=c_uint32, …)`
lookup	`map.lookup(&key)` → `u64*`	`map.lookup(key)` → value
update	`map.update(&key, &val)`	`map.update(key, val)`
emit	`events.perf_submit(ctx,&d,sizeof(d))`	`events.output(d)`
define event	`struct data_t {…}`	`@struct class Event: …`
userspace poll	`b.open_perf_buffer(cb)` + `perf_buffer_poll()`	`PerfEventArray::poll()`

What happens between start and stop?

1. Does BCC get 2 s before the trigger?
2. Does Python-BPF get 1 s before the trigger?
3. Do the 0.5 s and 2 s waits do any work?
4. Does the trigger run at the same speed for both programs?

Which program starts faster?

	BCC	Python-BPF
init CPU	0.4 s	0.02 s
child process	none	llc
header blocks	20,768	160

BCC compiles C at runtime. Python-BPF does not.

What do we run?

item	value
tracepoint	`sys_enter_execve`
trigger	2M `fork` + `execve` pairs
program	`map_1_perf_timed_count.py`
host	vmnono (3 CPUs)

Both frameworks run the same program.

What does the callback cost?

- count — prints one line
- empty — does nothing
- print — prints a constant

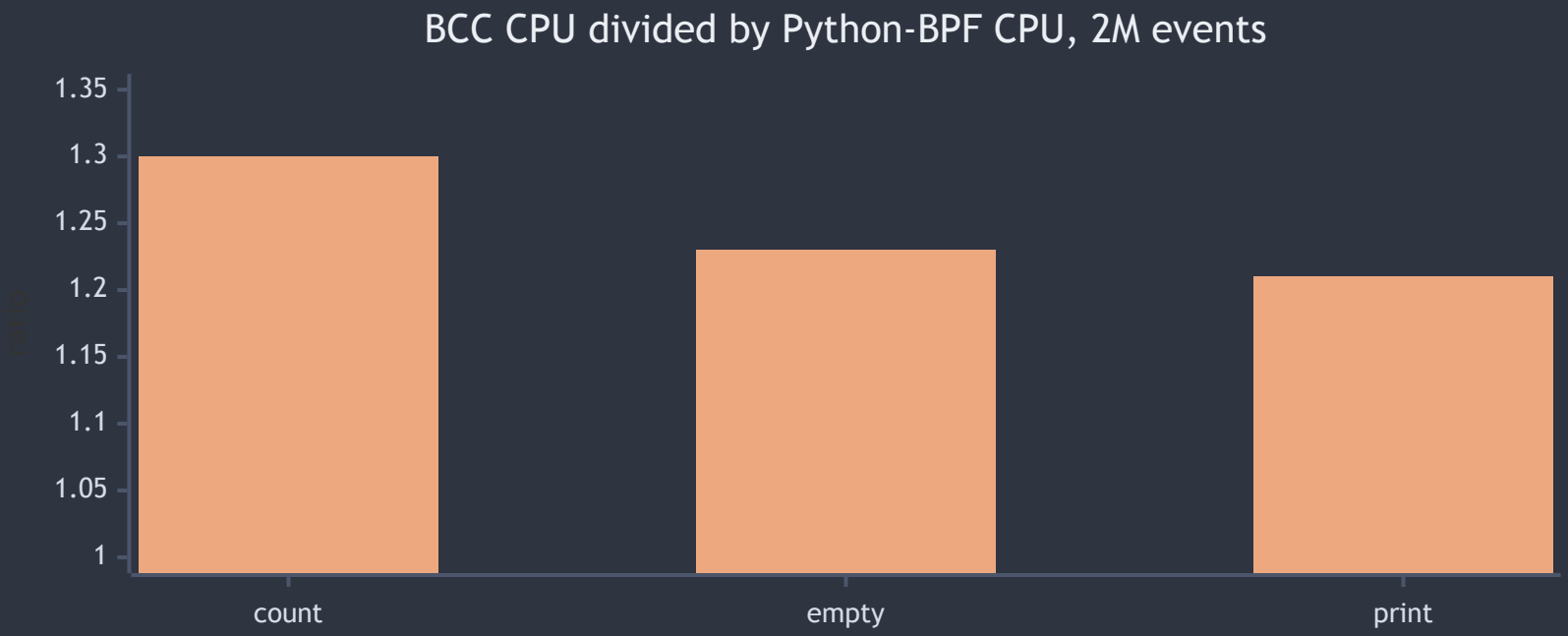
Same program. Only the callback changes.

Both frameworks deliver every event

	events delivered	trigger sent
BCC	2,000,000	2,000,000
Python-BPF	2,000,000	2,000,000

Both frameworks delivered 2,000,000 events. Nothing was lost.

Python-BPF uses less CPU



2M events, vmnono, un-profiled pass. BCC 134.6, 103.3 and 111.2 s. Python-BPF 103.7, 84.0 and 92.3 s.

An empty callback still costs more

callback body	BCC CPU	Python-BPF CPU	ratio
`pass`	103.3 s	84.0 s	1.23

The callback does nothing. BCC still spends 19.2 s more.

Where the CPU time goes

frame family, on the stack	BCC	Python-BPF
ctypes + libffi	84.0%	0%
clang	0.4%	0%
pybind11	0%	92.6%

`perf record`, 2M count arm. Share of sampled cycles with that frame anywhere on the stack.

BCC calls gettid once per event

syscall	BCC	Python-BPF
gettid	1,907,735	3
poll	1,978,114	1
total calls	3,913,286	2,055,624
calls per event	1.96	1.03

`strace -f -T`, 2M count arm. Every syscall counted. Counts only, not durations.

Wall time shows no difference

	BCC	Python-BPF	difference
wall	1763.6 s	1768.6 s	5 s, 0.3%
CPU	134.6 s	103.7 s	30.9 s, 30%

The trigger sets the pace, not the framework.

Three ways to get this wrong

trap	what it does	evidence
a long drain	adds CPU that is not the event	`T=15` at N=2000 fell to `T=0.5`
the profiler	roughly doubles the CPU it measures	134.6 s becomes 242.9 s
small N	measures startup, not per-event cost	2.1x at N=2000, 1.30x at 2M

Every number in this talk is CPU time, at 2M events.

Limitations

Seven gaps in Python-BPF, and one open question.

Limitation 1 — Python-BPF cannot deny a syscall

mechanism	Python-BPF
BPF-LSM program	no
`bpf_override_return`	no
`bpf_send_signal`	no

Enforcement runs outside the program.

Limitation 2 — A kill is not a block

stop method	when it acts
userspace kill	after the syscall
seccomp filter	before it

The vulnerable path already ran.

Limitation 3 — Two sections, one register struct

`struct_pt_regs` contains union members. Two sections that share it break the debug metadata.

sections with `struct_pt_regs`	result
one	compiles
two	debug metadata recurses

`RecursionError` in `function_debug_info.py:73`.

Workaround: one section per file. `kprobe` and `kretprobe` cannot share a file.

Limitation 4 — Python-BPF cannot read a kernel string

`ctx.si` holds a `struct filename *`. Three attempts to read its name:

attempt	helper	result
read `struct filename.name`	`probe_read`	`Struct type … not found`
read into a local `str(256)`	`probe_read_str`	`Variable 'buf' not found`
read into `c_uint64`	`probe_read_str`	8 bytes — truncates

What works reads the pointer, not the string: `probe_read(name_str, 8, fname_ptr)` → `0×ffff8f2d4705c000`, a valid kernel address.

Not garbage — the second dereference is missing. `probe_read` gives an `i64`; `probe_read_str` wants an `i8*`. No `inttoptr` between them.

Limitation 5 — Two gaps in the translator

construct	what happens
`ctx.args[0]`	falls through, no code emitted
`count += 1`	a plain add, not atomic

A map counter can lose updates.

Any other limitation?

What else have you hit with Python-BPF?

Is this Python-BPF, or the kernel?

program shape	trace output
one `print`, three values	works
four or more values in one call	silent

Both frameworks. It is a kernel rule.

`bpf_trace_printk` takes three values after the format string.

The count is not the finding. The silence is.

Python-BPF loads, attaches, drops the line, and says nothing.

Where each framework wins

axis	BCC	Python-BPF	why
startup CPU	0.45 s	0.02 s	BCC compiles C at runtime
per-event CPU	134.6 s	103.7 s	ctypes and clang cost cycles
syscalls per event	1.96	1.03	one gettid per event in BCC
peak RSS	175 MB	22 MB	clang loaded in the BCC process

What to choose

pick Python-BPF	pick BCC
startup: 0.02 s against 0.45 s	a struct as a map value works today
CPU at 2M: 1.30x lower	
syscalls: 1.03 against 1.96 per event	
memory: 22 MB against 175 MB	

Boundary: Python-BPF cannot deny a syscall in the kernel.

Q&A

Slides, dataset and scripts: github.com/nizarakbarm/pycon-tw-2026-bcc-vs-pythonbpf

Python-BPF: github.com/python-bpf/python-bpf · BCC: github.com/iovisor/bcc